

Chapter 6: Why hypothesis testing?



You are accused of a murder...

When will you be labelled as “guilty”?

🏴In a civilian court:

- 🏴It must be proven that you did commit the murder before you are labelled as guilty.
- 🏴Underlying assumption: “You are NOT GUILTY”.
- 🏴The judge has to decide your innocence or guilt on the basis of incomplete evidence.

In statistics terminology:

Null hypothesis $\rightarrow H_0$: You are NOT GUILTY

Alternative hypothesis $\rightarrow H_1$: You are GUILTY

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What are the possible errors and their consequences?

Analysis of hypotheses testing

Two possible conclusions from hypotheses-testing analysis:

- reject H_0
- fail to reject H_0

	The truth about the murder	
	H_0 : True (You are not guilty)	H_0 : False (You are guilty)
Based upon the sample evidence judge decides to		
Do not reject H_0	✓	Type-II
Reject H_0	Type-I	✓

What are the social costs of Type-1 and Type-2 error?

Analysis of hypotheses testing

The null hypothesis H_0 : conservative

- Type-1 error: reject H_0 when H_0 is true.
- $P(\text{reject } H_0 \text{ when } H_0 \text{ is true})$ is controlled directly.

Rejecting H_0 is a strong conclusion.

The alternative hypothesis H_a :

- Type-2 error: do not reject H_0 when H_a is true.
- $P(\text{do not reject } H_0 \text{ when } H_0 \text{ is not true})$ depends on the true value of the parameter and the sample size.

Accepting H_0 is a weak conclusion.

Fail to reject H_0 *or*
Cannot reject H_0 *or*
Do not reject H_0

Are Maine's lakes above the safety limits?

Federal law: The mean mercury level should be below 1.0 ppm

State law: The mean mercury level should be below 0.43 ppm

If you were the decision maker, how would you go about “issuing the health advisory??”

🦀 Formulate a hypothesis testing to decide whether there is enough evidence for the Maine fish to be poisonous.

🦀 Two possible actions:

🦀 To issue a health advisory.

Result of the hypothesis that
the Maine fish are poisonous.

🦀 Not to issue a health advisory.

Result of the hypothesis that
the Maine fish are not poisonous.

If the state of Maine is very concerned about the wealth of the fisheries, or it prefers to keep the status quo, how would you choose hypotheses of “the fish”?

🦞 Conservative hypothesis:

The Maine fish are NOT poisonous.

🦞 How to state this hypothesis in statistics?

🦞 $\mu = 0.43\text{ppm}$

🦞 What is the contradictory hypothesis?

🦞 $\mu > 0.43\text{ppm}$

$$H_0: \mu = 0.43$$

$$H_a: \mu > 0.43$$

If the state of Maine is very concerned about the health of the citizens, how would you choose hypotheses of “the fish”?

🦞 Conservative hypothesis:

The Maine fish are poisonous.

🦞 How to state this hypothesis in statistics?

🦞 $\mu = 0.43\text{ppm}$

🦞 What is the contradictory hypothesis?

🦞 $\mu < 0.43\text{ppm}$

$$H_0: \mu = 0.43$$

$$H_a: \mu < 0.43$$

Statistical hypotheses

A statistical hypothesis or hypothesis is a claim about the value of a single population characteristic or several population characteristics.

Definition: The null hypothesis denoted by H_0 , is the claim about one or more population characteristics that is initially assumed to be true. The alternative hypothesis, H_1 or H_a , is the assertion contradictory to H_0 .

Example: Milemaster tires

Strategy of Milemaster company: To introduce a new tire only if the true average life of the proposed tire is greater than 47,000 miles.

50 sample tires are produced.

Through accelerated life tests:

$$\bar{x} = 47,351 \text{ miles} \quad s = 1924.56 \text{ miles}$$

It is believed that X_i 's are normal random variables.

Question: To market or not to market this new tire

Example: Milemaster tires

μ : true average life of the proposed tire

Null hypothesis

$$H_0: \mu \leq 47000$$

Null action

Do not market the tire

Alternative hypothesis

$$H_a: \mu > 47000$$

Alternative action

Market the tire

H_0 is conservative.

Example: Milemaster tires

	The truth about the population mean tire life	
Based upon the sample evidence you decide to	H_0 : True (47,000 miles or under)	H_0 : False (over 47,000 miles)
Do not reject H_0	Correct decision	Type II error
Reject H_0	Type I error	Correct decision

Example: Milemaster tires

Consequences of errors??

Type-I error: Introduce the tire even if $\mu \leq 47,000$.

Very costly

Type-II error: Do not introduce the tire even if $\mu > 47,000$.

Depends on the real value of μ :

If $\mu = 47,500$: low cost

If $\mu = 57,000$: high cost of opportunity

Example: Milemaster tires

How to decide which hypothesis to believe?

$$H_0: \mu \leq 47,000 \quad \rightarrow \quad H_0: \mu = 47,000$$

$$H_a: \mu > 47,000 \quad \quad \quad H_a: \mu > 47,000$$

Reject H_0 if $\bar{X}$ is large enough.

What is large enough?

Example: Milemaster tires

Type I is serious, so we can set a bound on probability of type-I error.
Using this bound we can find the value of “large enough” to reject H_0 .

α : significance level

$$\begin{aligned}\alpha &= P\{\text{Type I}\} = P\{\text{reject } H_0 \text{ when } H_0 \text{ is true}\} \\ &= P\{\bar{X} > k, \mu = 47000\} = P\left\{\frac{\bar{X} - \mu}{S/\sqrt{n}} > \frac{k - \mu_0}{S/\sqrt{n}}\right\} \\ &= P\left\{Z > \frac{k - \mu_0}{S/\sqrt{n}}\right\} = P\left\{Z > \frac{k - 47000}{1924.56/\sqrt{50}}\right\} = P\{Z > z_\alpha\}\end{aligned}$$

Let $\alpha = 0.05$. Do we reject the null hypothesis for $\bar{x} = 47,351$ miles?

Testing population mean of a normal r.v. with large sample

$$H_0: \mu = \mu_0$$

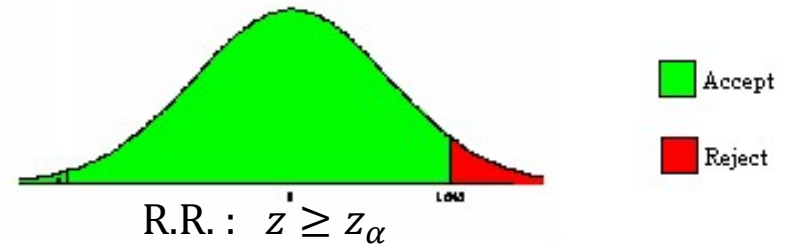
$$H_a: \mu > \mu_0$$

Reject H_0 if $\bar{X}$
is large enough.

$$Z = \frac{\bar{X} - \mu}{S/\sqrt{n}} \geq z_\alpha \Leftrightarrow \bar{X} \geq \mu + z_\alpha \frac{S}{\sqrt{n}}$$

Test statistics:

$$Z = \frac{\bar{X} - \mu}{S/\sqrt{n}}$$



Example: Milemaster tires

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Let $\alpha = 0.05$. Do we reject the null hypothesis for $\bar{x} = 47,351$ miles?

Relation between α and k

Type I is serious, so we can set a bound on probability of type-I error.

Using this bound we can find the value of “large enough” to reject H_0 .

α : significance level

$$\alpha = P\{\text{Type I}\} = P\{\text{reject } H_0 \text{ when } H_0 \text{ is true}\}$$

$$\frac{k - \mu_0}{S/\sqrt{n}} = z_\alpha \Rightarrow k = \mu_0 + z_\alpha \frac{S}{\sqrt{n}}$$

Example: Milemaster tires

We know the probability of type-I error. How about probability of type-II error?

$$\beta = P\{\text{Type II}\} = P\{\text{cannot reject } H_0 \text{ when } H_0 \text{ is false}\}$$

β can be calculated only at a given value of μ s.t. $\mu > 47,000$.

$$\beta = P\{\bar{X} < k, \mu = 47500\} = P\left\{\frac{\bar{X} - \mu_1}{S/\sqrt{n}} < \frac{k - \mu_1}{S/\sqrt{n}}\right\} = P\left\{Z < \frac{k - \mu_1}{S/\sqrt{n}}\right\} = \dots$$

$$\beta = P\{\bar{X} < k, \mu = 57000\} = \dots$$

Tests about a population mean

Upper tailed test

$$H_0: \mu = \mu_0$$

$$H_a: \mu > \mu_0$$

Lower tailed test

$$H_0: \mu = \mu_0$$

$$H_a: \mu < \mu_0$$

Two- tailed test

$$H_0: \mu = \mu_0$$

$$H_a: \mu \neq \mu_0$$

An estimator for $\mu = ?$

$$\bar{X}$$

How to use $\bar{X}$ to conclude about these tests?

$$\bar{X} > k_U$$

$$\bar{X} < k_L$$

$$\begin{aligned} \bar{X} &< k'_L \\ \bar{X} &> k'_U \end{aligned}$$

Testing population mean of a normal r.v. with large sample

$$H_0: \mu = \mu_0$$

$$\text{Test statistics: } Z = \frac{\bar{X} - \mu_0}{S/\sqrt{n}}$$

Test statistics is given under the assumption that H_0 is true.

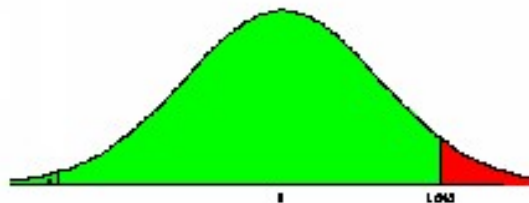
$$H_1: \mu > \mu_0$$

$$H_1: \mu < \mu_0$$

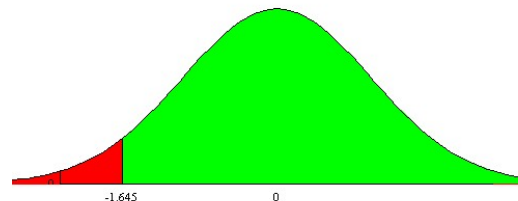
$$H_1: \mu \neq \mu_0$$

Accept

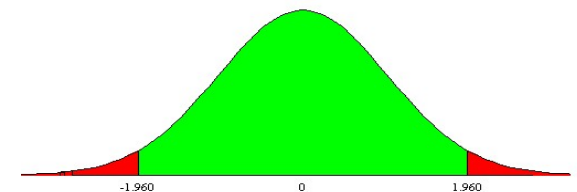
Reject



$$\text{R.R. : } z \geq z_\alpha$$



$$\text{R.R. : } z \leq -z_\alpha$$



$$\begin{aligned} \text{R.R. : } z &\leq -z_{\alpha/2} \\ \text{or } z &\geq z_{\alpha/2} \end{aligned}$$

Tests about a population mean of a normal r.v. with known variance

$$H_0: \mu = \mu_0$$

$$\text{Test statistics: } Z = \frac{\bar{X} - \mu_0}{\sigma/\sqrt{n}}$$

Test statistics is given under the assumption that H_0 is true.

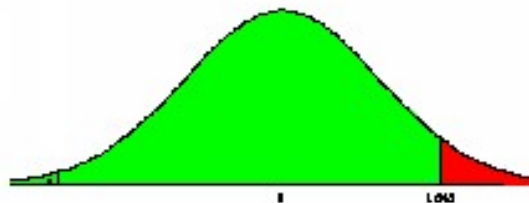
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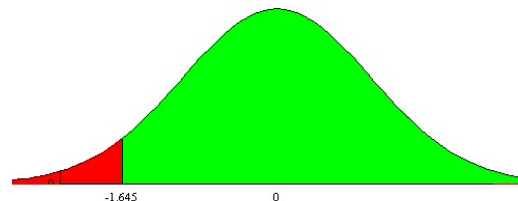
$$H_1: \mu \neq \mu_0$$

Accept

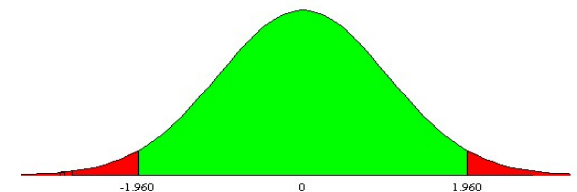
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